

EE5239 Homework 4

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1 Problem 1

1.1 2.1.6

Problem 2.1.6

It's equivalent to the original problem to maximize its logarithmic transformation, which is:

$$f(x) = \sum_{i=1}^n a_i \ln(x_i).$$

Define the Lagrangian $\mathcal{L}(x, \lambda)$ as:

$$\mathcal{L}(x, \lambda) = \sum_{i=1}^n a_i \ln(x_i) + \lambda \left(1 - \sum_{i=1}^n x_i \right).$$

- For each x_i :

$$\frac{\partial \mathcal{L}}{\partial x_i} = \frac{a_i}{x_i} - \lambda = 0 \Rightarrow x_i = \frac{a_i}{\lambda}.$$

and

$$\frac{\partial \mathcal{L}}{\partial \lambda} = 1 - \sum_{i=1}^n x_i = 0.$$

Substitute $x_i = \frac{a_i}{\lambda}$ into the constraint:

$$\sum_{i=1}^n \frac{a_i}{\lambda} = 1 \Rightarrow \lambda = \sum_{i=1}^n a_i.$$

Therefore, $x_i = \frac{a_i}{\sum_{j=1}^n a_j}$ for $i = 1, \dots, n$.

The unique solution is:

$$x_i = \frac{a_i}{\sum_{j=1}^n a_j}, \quad i = 1, \dots, n.$$

1.2 2.1.12 a

The algorithm is as follows:

Sort z from largest to smallest

Compute the threshold parameter $= (\sum_{j=1}^k z_{(j)} - 1)/k$ where k is the maximum integer where $z_k - \theta \geq 0$.

Compute the projection x as $x_i = \max(z_i - \theta, 0)$ for each i

2.3.1

We want to minimize the following:

$$\min_{x \in X} \left\{ \nabla f(x^k)^T (x - x^k) + \frac{1}{2s^k} (x - x^k)^T H^k (x - x^k) \right\}.$$

This can be rewritten as:

$$\nabla f(x^k)^T (x - x^k) + \frac{1}{2s^k} (x - x^k)^T H^k (x - x^k) = \frac{1}{2} \|x - (x^k - s^k (H^k)^{-1} \nabla f(x^k))\|_{\mu_k}^2 + \text{constant},$$

where the term $x^k - s^k (H^k)^{-1} \nabla f(x^k)$ is a scaled gradient step adjusted by H^k , and

$$\|z\|_{\mu_k} = \sqrt{z^T H^k z}$$

is the norm induced by H^k .

This is equivalent to

$$\frac{1}{2} \|x - (x^k - s^k (H^k)^{-1} \nabla f(x^k))\|_{\mu_k}^2 \quad \text{subject to } x \in X.$$

1.3 3.1.1

a. Lagrangian:

$$\mathcal{L}(x, \lambda) = x_1^2 + x_2^2 + \dots + x_n^2 + \lambda \left(\sum_{i=1}^n x_i - 1 \right)$$

KKT conditions: 1. $\frac{\partial \mathcal{L}}{\partial x_i} = 2x_i + \lambda = 0$ for $i = 1, \dots, n$ 2. $\frac{\partial \mathcal{L}}{\partial \lambda} = \sum_{i=1}^n x_i - 1 = 0$

From $2x_i + \lambda = 0$, we find $x_i = -\frac{\lambda}{2}$ So,

$$n \left(-\frac{\lambda}{2} \right) = 1 \Rightarrow \lambda = -\frac{2}{n}$$

Given $\lambda = -\frac{2}{n}$ the solution is $x_i = \frac{1}{n}$

Final solution:

$$x_i = \frac{1}{n}, \quad i = 1, \dots, n$$

b. Lagrangian:

$$\mathcal{L}(x, \lambda) = \sum_{i=1}^n x_i + \lambda \left(\sum_{i=1}^n x_i^2 - 1 \right)$$

KKT conditions: 1. $\frac{\partial \mathcal{L}}{\partial x_i} = 1 + 2\lambda x_i = 0$ for $i = 1, \dots, n$ 2. $\frac{\partial \mathcal{L}}{\partial \lambda} = \sum_{i=1}^n x_i^2 - 1 = 0$

From $1 + 2\lambda x_i = 0$, we find $x_i = -\frac{1}{2\lambda}$ So,

$$n \left(-\frac{1}{2\lambda} \right)^2 = 1 \Rightarrow \lambda = -\frac{1}{2\sqrt{n}}$$

Given $\lambda = -\frac{1}{2\sqrt{n}}$, the solution is $x_i = \frac{1}{\sqrt{n}}$

Final solution:

$$x_i = \frac{1}{\sqrt{n}}, \quad i = 1, \dots, n$$

c. Lagrangian:

$$\mathcal{L}(x, \lambda) = x^T x + \lambda(x^T Q x - 1)$$

KKT conditions: 1. $\frac{\partial \mathcal{L}}{\partial x} = 2x + 2\lambda Qx = 0$ 2. $\frac{\partial \mathcal{L}}{\partial \lambda} = x^T Q x - 1 = 0$

From $2x + 2\lambda Qx = 0$, we find $(I + \lambda Q)x = 0$ To satisfy $x^T Q x = 1$, we solve for the eigenvalue λ such that x is an eigenvector of Q .

Final solution:

$$x = \text{eigenvector of } Q \text{ with } x^T Q x = 1$$

2 Problem 2

I do not have the third edition of the textbook. I'm assuming these problems are the same as those from problem 1 but in the third edition instead of the second edition.

3 Problem 3

I chose option 2

4 Problem 4

The data consists of two features, symmetry and intensity, used to classify digits "1" (label 1) and "5" (label 0). We applied three models: linear regression, logistic regression, and Perceptron.

Linear regression was adapted for classification by setting a threshold on predictions. Using gradient descent with a constant step size of 0.01, the model minimizes squared errors. Logistic regression uses a sigmoid function to model class probabilities. We used gradient descent with the Armijo rule for adaptive step sizes, which dynamically reduces the step size if an update does not sufficiently decrease the error. This strategy prevents overly large or overly small steps, improving stability and speed.

The Perceptron adjusts weights only for misclassified points, moving the decision boundary with each update. Using a fixed learning rate, it stops when all points are correctly classified or after a maximum number of iterations.

All three of these models achieved very good accuracy as shown in figures below.

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Linear Regression - Training Accuracy: 99.42%
Linear Regression - Testing Accuracy: 98.11%
Logistic Regression - Training Accuracy: 99.55%
Logistic Regression - Testing Accuracy: 98.11%
Perceptron - Training Accuracy: 99.42%
Perceptron - Testing Accuracy: 98.35%
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